

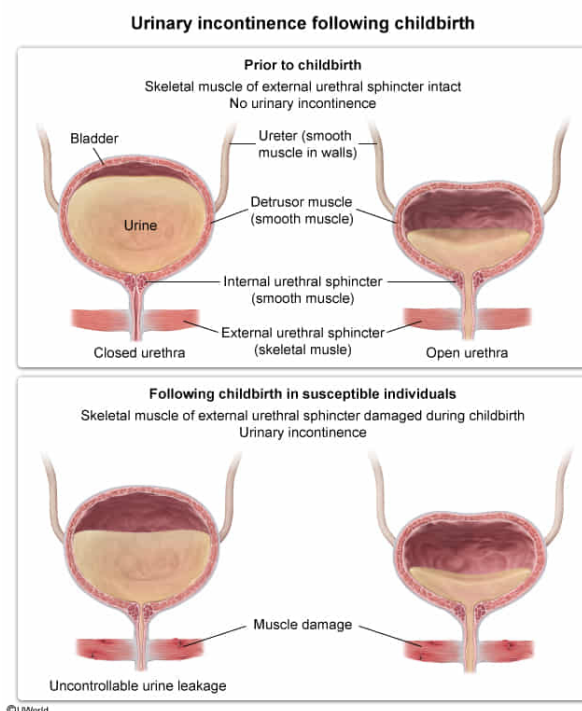
After giving birth, some females experience urinary incontinence, or the uncontrollable loss of urine from the body. Presumably, this occurs because childbirth damages the:

- ✔ ☒ A. skeletal muscle of the external urethral sphincter. [45%]
- ☐ B. smooth muscle of the external urethral sphincter. [12%]
- ☐ C. skeletal muscle of the internal urethral sphincter. [5%]
- ☐ D. smooth muscle of the internal urethral sphincter. [36%]

Correct

46% answered correctly

Explanation:



In the **excretory system**, **urine** is produced in the **nephrons** within each **kidney**. The urine from the kidneys is then funneled into each of the two **ureters**, tubes connecting the kidneys to the **bladder**. The process of urine exiting the body is known as **urination**.

Urination is under the control of both **smooth muscle** and skeletal muscle along the **urinary tract** (the system for removing urine from the body). Specifically, a layer of smooth muscle lining the bladder known as the **detrusor muscle** is relaxed while urine is stored in the bladder. During this storage process, two sphincter muscles are contracted so that urine cannot exit the body:

- The proximal sphincter muscle is known as the **internal urethral sphincter (IUS)**. The IUS is composed of smooth muscle and is under involuntary control by the **autonomic nervous system**.

- The distal sphincter, or the **external urethral sphincter (EUS)**, is composed of skeletal muscle and is under voluntary control by the somatic nervous system.

During urination, the detrusor muscle in the bladder contracts to push urine out and the IUS then relaxes to allow urine through. For urine to be excreted from the body, IUS relaxation must be paired with voluntary relaxation of the EUS. Relaxation of the IUS alone will not allow urine excretion.

The question states that some females experience an uncontrollable loss of urine from the body after giving birth known as urinary incontinence. The process of childbirth likely damages the sphincter that is under the female's **voluntary control**—the **EUS**, which is composed of **skeletal muscle**.

(Choice B) The EUS is composed of skeletal (*not* smooth) muscle.

(Choices C and D) Damage to the involuntary IUS *alone* would *not* cause the type of urinary incontinence described in the question. The voluntary EUS would *also* have to be damaged to allow uncontrollable urine loss from the body.

Educational objective:

Urine produced in the kidneys flows from the ureters to the bladder, where it is stored until its release. During urination, the detrusor muscle lining the bladder contracts while the internal urethral sphincter (under involuntary control) and the external urethral sphincter (under voluntary control) relax.

Time spent:0 sec

Subject:
Biology

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System:
Digestion and Excretion